

Blending

Problem

- Find the optimal purchasing and sales policy.

Sets:

- Oils: $o = 1(VEG1), 2(VEG2), 3(OIL1), 4(OIL2), 5(OIL3)$.

Parameters

- Sales price of final product: $SellPrice$
- Purchase price of raw oil o : $Cost_o$
- Hardness of raw oil o : $Hardness_o$
- Lower hardness bound: $LowerHardness$
- Upper hardness bound: $UpperHardness$
- Refinery capacity for Vegetable oils: $RefCapVeg$
- Refinery capacity for Non-Vegetable oils: $RefCapNonVeg$

Decision variables

- Amount of oil o bought and used: $x_o \geq 0$

Model

Objective:

- Objective is profit to be maximized:

$$\sum_o SellPrice \cdot x_o - \sum_o Cost_o \cdot x_o$$

Constraints:

- Lower hardness:

$$\sum_o Hardness_o \cdot x_o \geq LowerHardness \cdot \sum_o x_o$$

- Upper hardness:

$$\sum_o Hardness_o \cdot x_o \leq UpperHardness \cdot \sum_o x_o$$

- Refinery capacity for vegetable oils:

$$\sum_{o=1}^2 x_o \leq RefCapVeg$$

- Refinery capacity for Non-Vegetable oils:

$$\sum_{o=3}^5 x_o \leq RefCapNonVeg$$

- Positive buying:

$$x_o \geq 0, \forall o$$

The full model in Julia/JuMP is available with the name

Blending.jl

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#####
# Blending Assignment, "Mathematical Programming Modelling" (42112)

#####
# Intro definitions
using JuMP
using HiGHS
#####

#####

# INDEXES
Oils = ["VEG1" "VEG2" "OIL1" "OIL2" "OIL3"]
O = length(Oils)

# DATA
```

```

Cost      = [110 120 130 110 115] # cost of buing the different oils
Hardness = [8.8 6.1 2.0 4.2 5.0] # hardness of the different oils
SellPrice = 150 # selling price for final product
RefCapVeg = 200
RefCapNonVeg = 250
LowerHardness = 3
UpperHardness = 6
*****

#*****
# Model
blending = Model(HiGHS.Optimizer)

# how much buy of different oils
@variable(blending, x[1:0] >= 0)

# Maximize profit
@objective(blending, Max,
            sum(SellPrice*x[o] for o=1:0) -
            sum(Cost[o]*x[o] for o=1:0))

# lower hardness
@constraint(blending,
            sum(Hardness[o]*x[o] for o=1:0) >=
            LowerHardness*sum(x[o] for o=1:0))

# upper hardness
@constraint(blending,
            sum(Hardness[o]*x[o] for o=1:0) <=
            UpperHardness*sum(x[o] for o=1:0))

# veg capacity
@constraint(blending,
            sum(x[o] for o=1:2) <= RefCapVeg)

# non-veg capacity
@constraint(blending,
            sum(x[o] for o=3:0) <= RefCapNonVeg)

print(blending)
*****

#*****
# solve

```

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optimize!(blending)
println("Termination status: $(termination_status(blending))")
#####

#####
println("-----");
if termination_status(blending) == MOI.OPTIMAL
    println("RESULTS:")
    println("Blending result: $(objective_value(blending))")
    for o = 1:O
        println(" bought oil: $(value(x[o]))")
    end
else
    println(" No solution")
end
println("-----");
#####

#####
println("Successfull end of $(PROGRAM_FILE)")
#####

```

Comments to the Julia/JuMP program:

- Notice: For the lower- and upper hardness constraints on the final product we cannot have the hardness on the left-hand-side because it would involve dividing decision variables with other decision variables when we divide by the volume of the final product. What can be done to linearize this, is to multiply both sides of the constraints with the full volume leading to the constraints shown here, which are still valid but very importantly: Linear.